

Chapter 49: Green Nanofertilizer: Sustainable Nanotechnology for Precision Plant Nutrition

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Abstract

Major advances in agricultural productivity during the twentieth century have been made possible by conventional chemical fertilizers, but they also have a number of drawbacks, including low nutrient use efficiency, high production cost, and significant external effects like eutrophication, greenhouse-gas emissions, and soil degradation. Nanotechnology provides a pathway for reformulating plant nutrients at a nanoscale to uptake the nutrients better and release them in a controlled manner. The process of synthesizing nanoparticles with the help of plant extract, microorganisms, or any other biological or biowaste materials instead of toxic chemical reductants and solvents is called "green nanofertilizers" (GNFs). This chapter gives a systematic review of the rationale, synthesis pathways, classification, mode of action, characterization techniques, application methods, and field evidence for green nanofertilizers and explores the environmental safety, regulatory status, existing limitations, and future direction of the green nanofertilizers. The objective is to deliver a structured and up-to-date understanding in the field of nanotechnology, green chemistry, and sustainable agriculture.

Keywords: Green Synthesis, Nanofertilizer, Sustainable Agriculture, Fertigation, Phytotoxicity.

1. Introduction

1.1 Global Challenge for Fertilizer

A substantial amount of food would be required to feed a global population of 9.7 billion by mid-century with a limited supply of land and water. Despite playing a major role in this productivity since the advent of the Green Revolution in 1960, the usage efficiency of synthetic nitrogen, phosphorus, and potassium (NPK) fertilizers has not improved. The nutrient-use efficiency (NUE) of nitrogen in conventional fertilizers is generally reported to be 30–35%, and even less of the applied phosphorus is available to the crop, lost through volatilization, leaching, surface runoff, or fixation into unavailable forms of soil nitrogen (Jakhar et al., 2022). These losses not only have economic costs, but they are also a major contributor to eutrophication of waterways, nitrous oxide emissions, contaminated groundwater, and soil degradation over time (Nongbetet al., 2022). The micronutrient fertilizers, like the elements of zinc, iron, copper, manganese, boron, and molybdenum, are more prone to efficiency issues, as they are required in very low concentrations and easily form plant-unavailable complexes in the soil matrix (Yadav et al., 2023). The net result of global agricultural system relies on increasing amounts of fertilizer to maintain crop yields, where benefits are increasingly marginal and costs are growing ecologically.

1.2 Nanotechnology as a Solution

Nanotechnology, the ability to manipulate matter at the 1–100 nanometer scale, provides an alternative method to nutrient delivery based on structure. At this scale, materials have a dramatically increased surface-area-to-volume ratio, changed surface reactivity, and engineered porosity and coating, which can be finely tuned to release nutrients slowly and in response to plant demand (Raliya et al., 2018). The potential advantage of encapsulating or synthesizing the nutrient as a nanoparticle, nanocomposite, or nanocarrier is that it can allow a higher proportion of the applied nutrient to be available to the plant,

decrease the amount of fertilizer needed, and minimize the loss of nutrients to the environment from conventional granular or soluble fertilizer. Nutrient-use efficiency improvements reported for nanoformulations vary significantly based on nutrient, crop, formulation, and delivery systems, with reductions of nitrogen leaching and losses of nitrogen in runoff reported (Kah et al., 2019; Mim et al., 2025).

1.3 Importance of Green Nanotechnology

The previous synthesis protocols for nanoparticles were physical ones (ball milling, laser ablation, and vapor deposition) or chemical ones (chemical reduction, sol-gel, and co-precipitation), which require a lot of energy and hazardous reducing and capping agents like sodium borohydride, hydrazine, and organic solvents (Reverberi et al., 2019; Wu et al., 2022). These residues may remain on the surface of the nanotechnology product or enter the soil and water bodies, thereby casting doubt on the ecological benefits of the use of nanotechnology in the production of fertilizers even from the beginning. In the green synthesis approach, harmful chemicals are substituted by biological resources like plant extracts, bacteria, fungi, algae, yeast, agricultural wastes, and other biowaste, which provide various reducing, capping, and stabilizing agents such as polyphenols, flavonoids, terpenoids, proteins, and organic acids (Ovais et al., 2018; Thakur et al., 2025). This yields a nanoparticle is more cost-effective, less toxic, and often has a bioactive potential that supplies stress alleviation or growth-promoting attributes to the plant (Biswas et al., 2024). This property of the delivery of nutrients at the nanoscale level, coupled with an environmentally friendly approach, makes the “green nanofertilizer” a unique category in the field of nanofertilizers (Gupta, 2026).

2. Conceptual Framework

2.1 Nanofertilizer

Any fertilizer formulation containing at least one part of the active nutrient carrier within the nanoscale (generally 1-100 nm) is considered a nanofertilizer (Ijaz et al., 2020). The nanofertilizers can be composed of:

- Nutrient nanoparticles: The nutrient element itself as the core of the nanoparticles, such as ZnO, Fe₃O₄, and CuO nanoparticles.
- Nanocomposite or nanocarrier systems: The conventional nutrient is loaded onto or encapsulated within a nanoscale carrier such as a substrate of chitosan, zeolite, biochar, mesoporous silica, or hydroxyapatite that controls the release of the nutrient (Kottegoda et al., 2017; Fellet et al., 2021).
- Nano additives: Nano materials that are added to fertilizer granules to alter their solubility and/or interaction with soil colloids or the integrity of the coating (Madzokere et al., 2021).

2.2 Classification of Green Nanofertilizers

Green nanofertilizers can be classified most suitably based on the nutrient(s) they supply or provide (Nongbet et al., 2022; Yadav et al., 2023).

Table 1: Classification of green nanofertilizers by nutrient category and typical biological precursor.

Category	Representative Elements/Compounds	Typical Biological Precursor	Reference
Macronutrient-based	N, P, K, Ca, Mg, S (e.g., nano-urea, nano-DAP,	Plant leaf/peel extracts, biochar,	Kottegoda et al., 2017; Abdel-Aziz et

Category	Representative Elements/Compounds	Typical Biological Precursor	Reference
	nano-hydroxyapatite, nano-CaCO ₃)	chitosan	al., 2021; Fellet et al., 2021
Micronutrient-based	Zn, Fe, Cu, Mn, B, Mo (e.g., ZnO, Fe ₃ O ₄ /FeO, CuO, MnO ₂ nanoparticles)	Leaf, fruit-peel, and seed extracts; algae; fungal filtrates	Biswas et al., 2024; Ukidave&Ingale, 2022; Anas&Quraishi, 2025
Organic/nano-biofertilizer	Nitrogen-fixing or phosphate-solubilizing microbes encapsulated at nanoscale	Bacterial/fungal cultures, biopolymer carriers	Bettencourt et al., 2020; Panichikkal et al., 2019
Nano-enabled carrier/composite	Nano-zeolite, nano-chitosan, nano-biochar, nano-clay loaded with nutrients	Agricultural residues, marine biowaste (chitin/chitosan)	Sharma et al., 2020; Jahangirian et al., 2020
Nano-enhanced/stress-mitigating	Silicon, selenium, and other beneficial-element nanoparticles	Plant extracts, agro-industrial waste	Singh et al., 2025; Liu et al., 2021

3. Green Synthesis Approaches

3.1 General Mechanism

The general steps involved in the green synthesis of nanoparticles from biological sources include the bio-reduction of a metal-salt precursor such as a zinc, iron, or copper salt dissolved in water, the nucleation of metal or metal-oxide clusters; the growth and aggregation to form nanoparticles; the nucleation of metal or metal-oxide clusters; the growth and aggregation to form nanoparticles; and the capping/stabilization of the particles, which occurs when an excess of biomolecules binds to the surface of the nanoparticles and inhibits further growth and agglomeration (Ovais et al., 2018; Wu et al., 2022). The yield, particle size, and shape of the products are affected by various reaction parameters, such as the ratio of extract to salt, pH, temperature, and reaction time, which need to be optimized for each plant or microbial source, most often through empirical trial and error (Ijaz et al., 2020).

3.2 Plant-Mediated

The most common green synthesis is plant-mediated synthesis. Aqueous extracts of leaves, fruit peels, seeds, roots, or whole plants are added to a solution of metal salt, and both phenolics and flavonoids, terpenoids, and proteins in the extract are used as reducing and capping agents (Ovais et al., 2018). Some examples include iron, phosphorus, zinc, and silicon nanoparticles that have been green-fabricated from the extracts of green tea (*Camellia sinensis*) leaf (Biswas et al., 2024), pomegranate peel and spent coffee ground extract (Abdelmigid et al., 2022), and coriander (*Coriandrum sativum*) leaf extract (Ukidave&Ingale, 2022); calcium, magnesium, and silicon nanoparticles (Singh et al., 2025); and

manganese dioxide nanoparticles (Anas&Quraishi, 2025) from various plant extracts and applied in nano-fertilizer formulations. This route has the advantage of valorizing agricultural residues that are otherwise discarded, is easy to scale up and perform in a lab environment, and does not need any special microbial infrastructure.

3.3 Microbial Synthesis

There are also other mechanisms that bacteria, fungi, yeast, and algae can use to reduce metal ions extracellularly with the help of enzymes and metabolites secreted from the cell or intracellularly with the help of uptake and enzymatic reduction in the cell (Koul et al., 2021; Jadoun et al., 2022). The use of fungal systems is more attractive as they produce high amounts of extracellular reducing enzymes, are well tolerant of relatively high metal ion concentrations, and are relatively easy to culture at large scale (BoroumandMoghaddam et al., 2015). Bacterially synthesized iron–manganese bimetallic nanoparticles are being used as biofertilizers for plants, with a bacterial auxin complex serving as a reducing and functionalizing agent (Bettencourt et al., 2020). Additionally, supplementing the plant-beneficial bacterium *Pseudomonas monteilii* with biogenic gold nanoparticles has enhanced its beneficial effects on plants (Bettencourt et al., 2020); the plant-beneficial bacterium *P.monteilii* has increased its beneficial effects on plants as a reducing and functionalizing agent (Bettencourt et al., 2020); and the supplementation of *P.monteilii* with biogenic gold nanoparticles increased its plant-beneficial effect (Panichikkal et al., 2019). Silicon- and calcium-based nanofertilizers are intriguing precursors, and among them, algal biomass, which is available in large quantities, is an emerging option that can be used as a by-product in aquaculture or wastewater treatment.

3.4 Biopolymer- and Biowaste-Derived Carrier Systems

Instead of the nutrient nanoparticle, the focus of a parallel green route is on the carrier, which regulates its release. Crustacean shell waste is used to develop chitosan-based nano fertilizer, which was proven to increase the photosynthesis of source tissues in maize (Sharma et al., 2020) and to develop nanofibers to change the morphology and metabolite content in roselle (Mirheidari et al., 2022). The pyrolyzed crop residues (biochar) and plant biomass (cellulose nanocrystals) are also utilized to make bio-nanocomposite carriers with conventional soluble nutrients such as urea, phosphate, potassium salts, or micronutrient salts, which are coated or cross-linked to prevent them from dissolving too quickly (Fellet et al., 2021). Similarly, low-impact, synthesized waste-based NPK nanofertilizers have been proven to improve the growth and yield of bell pepper (Abdel-Aziz et al., 2021), and zeolite/Fe₂O₃ synthesized by green methods was tested as a smart, low-toxicity iron nanofertilizer (Jahangirian et al., 2020).

4. Mechanisms of Action in Plants

The multiple mechanisms of action and the downstream effects on plants caused by green nanofertilizers are interconnected. The subsections below summarize and discuss these relationships.

4.1 Enhanced Nutrient Uptake

Nanoparticles have the potential to be taken up by plants via pathways that are frequently inaccessible to bulk fertilizer particles, such as the epidermal pores, stomata, and cuticularnanochannels, and some formulations can then be translocated into the phloem and xylem (Karny et al., 2018; Wang et al., 2023). Using *Perillafrutescens* as a model plant, Ha et al. (2021) have recently shown that foliar application of nanoparticles in a droplet on the plant surface can be adsorbed onto and penetrate through the stomatal pathway, which is a direct visual demonstration of foliar penetration of nanoparticles. Their high surface

reactivity also promotes the dissolution rate and ion availability in the rhizosphere compared to mineral sources, and cap biomolecules can chelate co-occurring soil nutrients, further promoting the absorption of the roots (Achari&Kowshik, 2018; Lv et al., 2019).

4.2 Controlled and Targeted Release

One of the unique functional benefits of green nanofertilizers based on nanocarriers is the release mechanism. Kottegoda and colleagues (2017) demonstrated that the release of nitrogen from urea–hydroxyapatite nanohybrids is significantly extended as compared to the release of N from conventional soluble urea, which was reasoned to be due to the high binding affinity between the hydroxyl groups of urea and the surface of the hydroxyapatite nanoparticles. Nanocarriers based on chitosan, zeolite, and biochar have also been documented to prolong the nutrient release period compared to conventional fertilizer to closely align nutrient uptake to the crop's development over a growing season and decrease the number of re-application rounds (Fellet et al., 2021; Madzokere et al., 2021). The controlled release effect is the main mechanism of action that nanofertilizers offer to minimize leaching and volatilization losses from the soil compared to broadcast application of conventional fertilizers (Liu & Lal, 2015).

4.3 Stress Alleviation

In addition to basic nutritional effects, certain classes of green-synthesized nanoparticles, such as silicon nanoparticles, selenium nanoparticles, calcium nanoparticles, and metal-oxide nanoparticles (MONOs), activate the physiological and biochemical pathways of stress response, which lead to an increase in the activity of antioxidant enzymes, osmosis adjustment, and photosynthetic efficiency and, consequently, to an improvement in drought, salinity, heat, and heavy metal tolerance in crops. Roosta et al. (2023) reported that MnO₂ nanoparticle application on the foliage changed the growth and physiological traits of pennyroyal (*Menthapulegium*) under various cultivation systems, which was equivalent to the application of nanofertilizer and remediation. Anas and Quraishi (2025) also reported that green-fabricated MnO₂ nanoparticles not only boosted antioxidant mechanisms in wheat but also reduced chromium toxicity. Rizwan et al. (2019) showed that foliar application of zinc oxide nanoparticles along with biochar reduced the Cd accumulation in maize plants in contaminated soil. Selenium nanoparticles are being proposed as a new way to deliver selenium fertilizer, which causes less disruption to soil microorganisms than using selenium salts (Liu et al., 2021).

4.4 Soil and Rhizosphere Interactions

There are also interactions between the nanofertilizers and the wider soil–plant–microbe system, such as the impact of the organic corona and the dissolution products on rhizosphere pH and organic matter content and the impact on the composition of soil microbes (Kalwani et al., 2022). Chitosan-based NPK nanocomposites have been reported to increase water-stable soil aggregate stability and hydraulic conductivity and decrease bulk density, which was partially attributed to the nanocarrier itself as organic matter. These same soil–nanoparticle interactions are also the reason for the legitimate ecological concerns at higher application rates.

5. Characterization of Green Nanofertilizers

To ensure quality control as well as to achieve correlation between the properties of nanoparticles and agronomic responses, rigorous physicochemical characterization is required (Ijaz et al., 2020; Jahangirian et al., 2020). These following methods are generally employed:

Table 1: Common physicochemical characterization techniques used for green nanofertilizers.

Technique	Information Obtained	Source(s)
UV–Visible spectroscopy	Confirms nanoparticle formation via characteristic surface plasmon resonance (metal NPs)	Ijaz et al., 2020
Dynamic light scattering (DLS)	Hydrodynamic particle size distribution, polydispersity, colloidal stability	Jahangirian et al., 2020
Zeta potential analysis	Surface charge and colloidal/storage stability	Jahangirian et al., 2020
Transmission/scanning electron microscopy (TEM/SEM)	Particle size, morphology, aggregation state	Abdelmigid et al., 2022
X-ray diffraction (XRD)	Crystalline phase and crystallite size	Abdelmigid et al., 2022
Fourier-transform infrared spectroscopy (FTIR)	Identification of capping biomolecules on the nanoparticle surface	Ovais et al., 2018
Energy-dispersive X-ray spectroscopy (EDX/EDS)	Elemental composition	Ijaz et al., 2020
Thermogravimetric analysis (TGA)	Quantification of organic coating content and thermal stability	Fellet et al., 2021

6. Application Methods in Agriculture

The methods for the delivery of green nanofertilizers are usually one or more of the following, depending on specific crops, growth stage, and nutrient objectives (Ding et al., 2023; Pereira et al., 2021):

- Foliar spraying: Spraying the surface of the leaves, effective for quick correction of micronutrient deficiencies and to deliver stress-alleviating nanoparticles, such as in legume crops by introducing calcium/magnesium/silicon nanofertilizers (Singh et al., 2025) and zinc oxide nanoparticles in tomato crops and lentil production (Ahmed et al., 2021; Kolencik et al., 2022).
- Soil application/mixing: Incorporation into the root zone, typically to release macronutrients over a given time period, e.g., season-long controlled release of macronutrients like urea–hydroxyapatite nanohybrids (Kottegoda et al., 2017).
- Fertigation: Involves fertigation through a drip or micro-irrigation system, which enables exact and repeatable dosing at the right time consistent with the requirements of the crop (Yadav et al., 2023).
- Seed priming/coating: Seed priming/coating (also using silver, silica, and zinc nanoparticles) has also been reported to improve germination vigor and early establishment of seedlings (Bayat et

al., 2021; Pereira et al., 2021), while a bacterially derived (Enterococcus-based) nanofertilizer applied at the seedling stage has also been noted to improve early growth rates and vigor in wheat (Batool et al., 2024).

7. Evidence of Agronomic Efficacy

In many greenhouse and field trials, the green-synthesized nanofertilizers and others have been linked to observed benefits over conventional fertilizers:

- Nitrogen-use efficiency: This increase is comparable to the increase in NUE that has been reported on nano-formulations that are biosynthesized and are reported to be 30–35% higher than the NUE of conventional nitrogen sources, which are reported to follow a controlled-release mechanism as described by Kottegoda et al. (2017) for the urea–hydroxyapatite nanohybrids.
- Yield outcomes: Wheat and rice treated with plant-derived metallic nanoparticles showed yield improvements of 10% and 8%, respectively, while legumes treated with bacterially derived nanofertilizer made of hydroxyapatite exhibited a 5% increase in germination and seedling vigor; additionally, pomegranate showed a 4% increase in fruit and growth parameters (Batool et al., 2024; Abdelmigid et al., 2022).
- Stress tolerance: Green nanoparticles, like silicon-based, calcium-based, and manganese-based, have shown their efficacy in enhancing salinity, drought, and heavy metal stress tolerance, such as alleviation of chromium toxicity in wheat (Anas&Quraishi, 2025) and alleviation of cadmium toxicity in maize (Rizwan et al., 2019).
- Soil physical properties: In saline soil, the water-stable soil aggregates, hydraulic conductivity, and porosity have been reported to be higher with the chitosan-based NPK nano-fertilizer applications as compared to the use of conventional fertilizer.
- Micronutrient biofortification: The use of foliar zinc oxide nanoparticles has been associated with better tomato production results, and when combined with plant growth–promoting bacteria in common bean (Jalal et al., 2023), it has been possible to achieve agronomic biofortification.

8. Environmental and Toxicological Considerations

Several categories of concern include:

- Soil accumulation and persistence: Long-term behavior in soil for metal oxide and carbon-based nano-formulations upon continuous application is not fully understood (Nowack&Bucheli, 2007; Bouhadi et al., 2025).
- Soil microbiome effects: Kalwani et al. (2022) reported that the toxicity of the nanofertilizer to soil microbiota depends on its concentration and reviewed the evidence that low to moderate concentrations are not toxic to microbial biomass but rather can be beneficial, while higher concentrations can be detrimental to microbial biomass and nutrient mineralization.
- Phytotoxicity and food-chain entry: Phytotoxicity can also occur with excessive concentrations of nanoparticles; for instance, engineered nanoparticles of titanium dioxide and molybdenum disulfide were found to affect the metabolic profile and physiological response of cucumber foliage (Huang et al., 2020).

9. Limitations

- Mechanistic uncertainty: Knowledge of the exact molecular mechanism involved in the reduction and capping of metal ions by plant or microbial biomolecules is incomplete for several species, which makes it difficult to design processes rationally.
- Scale-up and reproducibility: Optimizing a process for small-scale use does not always correspond to the natural variability in biological feedstocks in a large batch to yield predictable results in the laboratory.
- Limited long-term and multi-crop field validation: Despite promising efficacy data, many of these results are limited to pot or greenhouse trials or one-season field trials of a small variety of crops.

10. Future Perspectives

- Biowastevalorization: More application of agro-industrial wastes as low-cost, high-volume precursors like fruit peels, husks, spent coffee grounds, and crustacean shell wastes for chitosan and the production of nanofertilizers based on the principle of a circular economy.
- Multifunctional formulations: Developing new formulations of nanofertilizers to deliver more than a single input, such as nutrients, alleviating abiotic stress, and crop protection, such as dual nanofertilizer/remediation agents for contaminated soils, can minimize the number of agrochemical inputs.
- Precision and smart delivery: Precision agriculture with sensors, which allows the release of nutrients in the soil according to the actual soil and plant condition, not a preprogrammed release curve.
- Standardized testing and protocols: Standardization of characterization and field testing across the community to enable comparison of efficacy and safety data between studies and harmonization within regulation.
- Life-cycle and techno-economic assessment: Further studies exploring the full environmental and economic impacts of the production of green nanofertilizers compared to conventional fertilizer production to provide quantitative evidence for the sustainability benefits of green nanofertilizers.

11. Conclusion

Green nanofertilizers overlap three contemporary challenges in the agricultural sector: firstly, providing food for an increasing population on a limited land and nutrient basis; secondly, reducing the environmental side effects brought by conventional fertilizers; and thirdly, reducing the environmental side effects of the fertilizer application process. Green nanofertilizers represent a promising and ethically sound route to better use of plant nutrients, using biologically derived synthesis methods inspired by plants, microorganisms, and biowaste from agriculture as an alternative to the hazardous physical and chemical methods of nanoparticle synthesis. The results from all of these types of macronutrient, micronutrient, and composite-carrier formulations are so consistently good that they produce superior nutrient-use efficiency, lower environmental losses, and sometimes extra stress-tolerance and crop-protection benefits compared to conventional fertilizers. However, there are also limitations in the field with regard to reproducibility; the evidence base is still dominated by greenhouse data rather than long-term field data; the risks of ecotoxicological effects are not fully understood; and the regulatory apparatus is not well developed. A long-term interdisciplinary effort that combines green chemistry, soil science, plant physiology, ecotoxicology, and agricultural policy is currently in progress and is necessary to fully realize the potential of green nanofertilizers, but it has not yet been completed.

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